

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems. (L4, L5)

Assessment Tool: Viva Voce

Weightage: 10%

QUESTIONS:

Total Marks: 20

1. Define SOAP.
2. Define WSDL.
3. What is an API?
4. Define XML Schema.
5. What is a Web Service?
6. Define REST.
7. What is a Protocol?
8. Define Endpoint in web services.
9. What is a Service in web applications?
10. What is the difference between SOAP and REST?
11. What is the role of WSDL in web services?
12. What is JSON and how is it used in APIs?
13. What is HTTP and why is it used in web services?
14. What is meant by request and response in web communication?
15. What is the importance of XML in web services?
16. What is the purpose of an API endpoint?
17. What is the difference between HTTP and HTTPS?
18. What is meant by RESTful services?

19. What is a namespace in XML?

20. What is the role of a client and server in web services?

Code of Conduct:

I P Madhavi/192372133 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Madhavi

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively

1. Define SOAP.

Answer: SOAP (Simple Object Access Protocol) is an XML-based communication protocol used to exchange structured information between applications over the internet. It provides secure and reliable communication between client and server.

2. Define WSDL.

Answer: WSDL (Web Services Description Language) is an XML document that describes a web service. It defines the operations, input/output messages, communication protocol, and service endpoint.

3. What is an API?

Answer: API (Application Programming Interface) is a set of rules that allows different software applications to communicate and exchange data with each other.

4. Define XML Schema.

Answer: XML Schema (XSD) is a language used to define the structure, elements, attributes, and data types of an XML document. It validates XML data.

5. What is a Web Service?

Answer: A web service is a software service that enables communication and data exchange between different applications over a network using standard protocols like HTTP, SOAP, or REST.

6. Define REST.

Answer: REST (Representational State Transfer) is an architectural style for developing web services that use HTTP methods such as GET, POST, PUT, and DELETE to exchange data, usually in JSON format.

7. What is a Protocol?

Answer: A protocol is a set of rules that defines how data is transmitted and communicated between devices or applications over a network.

8. Define Endpoint in Web Services.

Answer: An endpoint is the URL or address where a web service is available and receives client requests.

9. What is a Service in Web Applications?

Answer: A service is a software component that performs specific functions, such as processing requests, retrieving data, or providing business logic to clients.

10. What is the Difference Between SOAP and REST?

Answer:

SOAP	REST
Protocol.	Architectural style.
Uses XML only.	Uses JSON, XML, etc.
More secure and reliable.	Faster and lightweight.

11. What is the Role of WSDL in Web Services?

Answer: WSDL acts as a contract between the client and the web service. It describes the available operations, messages, protocols, and endpoint details.

12. What is JSON and How is it Used in APIs?

Answer: JSON (JavaScript Object Notation) is a lightweight data format used to exchange information between client and server in REST APIs. It is easy to read and process.

13. What is HTTP and Why is it Used in Web Services?

Answer: HTTP (Hypertext Transfer Protocol) is a communication protocol used to send requests and receive responses between clients and servers in web services.

14. What is Meant by Request and Response in Web Communication?

Answer:

- Request: Message sent by the client to the server asking for data or an action.
- Response: Message returned by the server containing the requested data or status.

15. What is the Importance of XML in Web Services?

Answer: XML is used to store and exchange structured data. It provides platform-independent communication and is the standard format used in SOAP web services.

16. What is the Purpose of an API Endpoint?

Answer: An API endpoint identifies the location of a specific API resource. Clients send requests to the endpoint to access or modify data.

17. What is the Difference Between HTTP and HTTPS?

Answer:

HTTP	HTTPS
-------------	--------------

No encryption.	Encrypted using SSL/TLS.
----------------	--------------------------

Less secure.	More secure.
--------------	--------------

Port 80.	Port 443.
----------	-----------

18. What is Meant by RESTful Services?

Answer: RESTful services are web services that follow REST principles and use HTTP methods (GET, POST, PUT, DELETE) to perform operations on resources.

19. What is a Namespace in XML?

Answer: A namespace is a unique identifier in XML that prevents naming conflicts between elements from different XML documents.

20. What is the Role of a Client and Server in Web Services?

Answer:

- Client: Sends requests to access web services and receives responses.
- Server: Processes client requests, performs operations, accesses databases if needed, and sends responses back to the client.